Joint Velocity Spectral Analysis and APS Compensation in Ground-Based SAR via *k*-Means Clustering

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***Abstract*—Interferometric ground-based synthetic aperture radars (GB-SARs) are widely used for their capability to measure small displacements with optimal illumination geometry and short revisit times. However, achieving high-precision displacement measurements requires removing phase disturbances, particularly those induced by variations in atmospheric refractivity, which manifest as an atmospheric phase screen (APS) in the interferometric images. Traditional APS estimation relies on model-based approaches, which assume the presence of motionless pixels and estimate APS through polynomial regression or spatial spectral analysis. These methods require that most of the scene remains stable and spatially homogeneous, which is often not the case. In this work, we propose a novel time-domain spectral analysis approach to estimate the average velocity of pixels exhibiting similar phase characteristics. The estimated velocity component is then removed in the complex domain, allowing APS evaluation over all pixels as if they were virtually motionless. Subsequently, APS is corrected in the original interferograms. To group pixels with similar phase behavior, a *k*-means clustering strategy is adopted. The proposed algorithm was assessed through simulations and validated with experimental datasets acquired in two distinct scenarios: an open-pit copper mine and a glacier.**

***Index Terms*—Atmospheric phase screen (APS), ground-based synthetic aperture radar (GB-SAR), interferometric GB-SAR, joint velocity spectral analysis, *k*-means clustering, time-series spectral analysis.**

I. Introduction

INTERFEROMETRIC ground-based synthetic aperture radar (GB-SAR) has been widely employed over the last two decades for monitoring diverse scenarios, including natural environments, infrastructures, dams, and open-pit mines [1]. The success of GB-SAR is related to its flexibility in revisit time and illumination angle [2], [3], which allows maximizing line-of-sight displacement detection and enabling the measurement of rapid surface deformations. By exploiting the interferometric phase (IP) between repeated acquisitions, GB-SAR can detect displacements at the submillimetric level. Achieving such performance, however, requires considering and correcting phase disturbances [4], [5], [6]. Among them, variations in atmospheric refractivity generate the so-called "atmospheric phase screen" (APS), which represents one of the main disturbance sources in GB-SAR applications.

Changes in atmospheric refractivity induce propagation delays in the radar signal [7]. In principle, the refractive index can be estimated from macroscopic meteorological parameters such as temperature, pressure, and humidity. Since the early development of radiometry, some semi-empirical models have been proposed to calculate the refractive index from such parameters [8], [9]. These models are accurate at a local scale, while failing to capture the spatially variable distribution of APS, which strongly depends on local weather conditions. For this reason, refractive index models are not reliable for APS correction in GB-SAR. Alternative approaches rely on auxiliary data, such as numerical weather prediction [10], [11] spectroradiometers and GPS observations [12]. Nevertheless, their limited spatial resolution prevents effective APS compensation in interferometric images. To overcome this, several authors have proposed model-based solutions tailored for GB-SAR without requiring external data [13], [14], [15], [16], [17].

Model-based approaches typically assume a homogeneous spatial distribution of refractive index, depending on the scenario. For flat terrain, a linear dependence on range is commonly assumed [13], [14]. For sloped or mountainous terrain, a vertically stratified model with horizontal homogeneity is often adopted. Consequently, APS is generally modeled as a polynomial function of range and elevation [7], [18]. Polynomial coefficients are estimated through multiple polynomial regression (MPR) using coherent pixels not affected by motion. Since MPR requires unwrapped interferograms, Falabella et al. [17] recently proposed a method for APS estimation directly in wrapped interferograms using Fourier transform analysis along range.

A limitation of model-based approaches is that they assume most of the image pixels remain stationary over time. While this assumption holds in many cases, it breaks down in highly dynamic scenarios, such as glaciers [19], [20], where many pixels undergo displacement. In such cases, APS evaluation becomes restricted to a few motionless pixels, and its reliability must be checked on a case-by-case basis.

In this work, we propose an algorithm that exploits the Fourier transform in the time domain to estimate the average velocity of each pixel to improve the APS compensation. The retrieved average velocity is mapped in the complex phase plane and subtracted from the original interferogram,

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producing an image where all pixels can be treated as a virtually motionless. APS of virtually motionless image can then be estimated using a model-based method and subsequently removed from the original interferogram to recover surface displacements. Compared with existing approaches, the proposed method introduces an additional step that maximizes the number of pixels contributing to MPR-APS. Moreover, by performing the analysis in the spectral domain, the method avoids phase-wrapping issues due to displacements larger than a quarter of a wavelength. The frequency domain of the Fourier transform is inversely proportional to revisit time, so wrap occurs for larger displacement speed.

Reliable spectral analysis requires a sufficiently high signal-to-noise ratio (SNR). This can be achieved either by extending the temporal dataset or by averaging pixels with similar phase behavior. Since long datasets risk coherence loss [5], the second approach has been preferred, and k-means clustering has been employed to group pixels with similar temporal phase signatures. The average velocity of each cluster is then computed by averaging their complex spectrograms. While clustering in GB-SAR applications is commonly used to isolate turbulent areas [21], [22], here it is used to maximize the number of pixels available for APS modeling. Other forms of pixel aggregation can also be used to increase the statistical number of scatterers contributing to the average. Methods such as distributed-scatterer (DS)–oriented techniques or multilook approaches, however, are typically based on amplitude information [23], [24], [25]. For this reason, they can be effectively applied only after identifying areas with homogeneous phase evolution through the k-means clustering stage. Naturally, the velocity retrieved in this way is an approximate estimate, and a further displacement analysis is required after APS compensation.

While the concept of using spectral analysis to characterize deformation signals is known in the general InSAR context, its application as a preprocessing step to enhance APS modeling for GB-SAR interferograms is relatively unexplored. The proposed method introduces a joint framework in which temporal spectral estimation and clustering are combined to derive a reliable velocity component that can be removed in the complex domain, allowing APS to be estimated over all pixels. This significantly extends the applicability of model-based APS approaches to scenarios where large areas are subject to displacement.

The remainder of this article is organized as follows. Section II introduces the signal model and describes the APS removal approach based on velocity estimation. Section III details the k-means clustering strategy adopted in the proposed method. Section IV reports results from simulations under realistic refractivity variations, while Section V presents the validation of the algorithm in two experimental scenarios: an open-pit copper mine in South America and an alpine glacier.

II. Signal Modeling and Atmospheric Correction

A. Interferometric Signal Model

A GB-SAR image can be represented as a complex matrix I, where the amplitude provides information on target positions within the scene, while the phase can be exploited to detect motion. In particular, the displacement can be retrieved through interferometry. The IP, ϕ, is defined as the argument of the complex coherence between two acquisitions at different times

$$\phi = \angle\left(I\left(t\right) \cdot I^{*}\left(t + \Delta t\right)\right) \tag{1}$$

where t and $t + \Delta t$ denote two observations.

The IP can be decomposed into three main contributions: the displacement term ϕ_{disp}, the APS term ϕ_{APS}, and the noise term ϕ_{noise}

$$\phi = \phi_{\text{disp}} + \phi_{\text{APS}} + \phi_{\text{noise}}. \tag{2}$$

Usually the noise term is negligible, so to correctly retrieve the displacement, it is essential to estimate and remove the APS. The APS contribution is approximately linear with respect to range, R

$$\phi_{\text{APS}} = \frac{4\pi}{\lambda}\left(\delta n_0 + \delta n_1 \cdot R + o\left(R^2\right)\right) \tag{3}$$

where δn_0 and δn_1 are the differences in refractive index between two acquisitions (constant term and range-dependent term, respectively), $o(R^2)$ is a possible residual with R^2 dependency that models the vertical stratification. Since δn_0 and δn_1 vary with atmospheric conditions, they can be treated as ergodic random variables.

The displacement phase is related to the range displacement ΔR as

$$\phi_{\text{disp}} = \frac{4\pi}{\lambda}\Delta R \tag{4}$$

where λ is the wavelength of the central frequency. As a first-order approximation, the displacement is assumed to vary linearly with time

$$\phi_{\text{disp}} = \frac{4\pi}{\lambda}\Delta R = \frac{4\pi}{\lambda}\left(v \cdot t + o\left(t^2\right)\right) \tag{5}$$

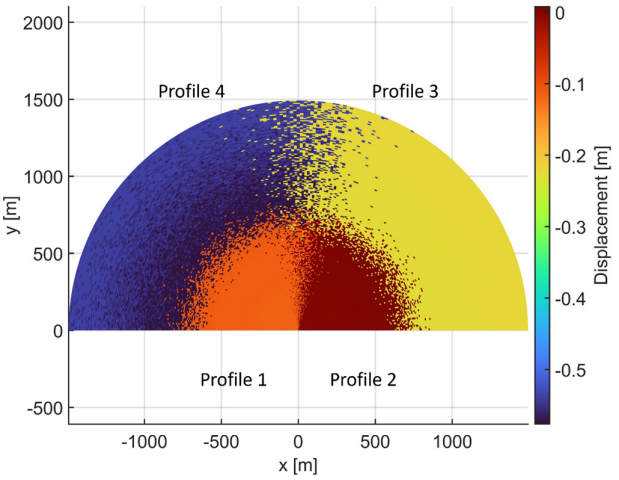


Fig. 1. Spatial distribution of four areas with different profiles.

TABLE I
Nominal Speeds of Simulated Clusters

Profile	Speed (m/s)
1	0
2	-0.56×10^{-5}
3	-1.11×10^{-5}
4	-2.78×10^{-5}

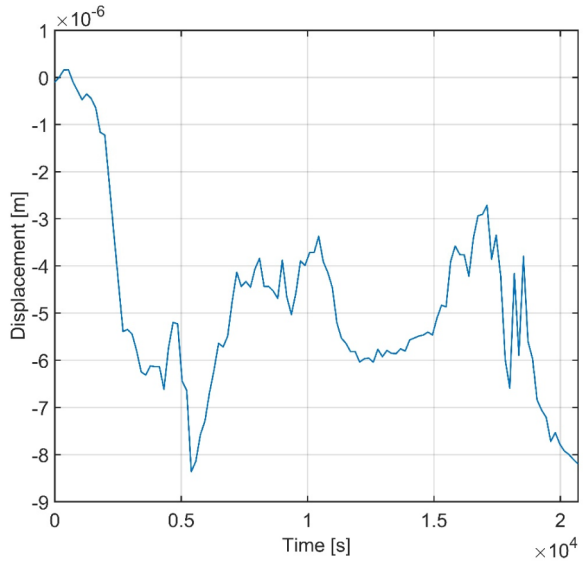


Fig. 2. APS phase used for simulation.

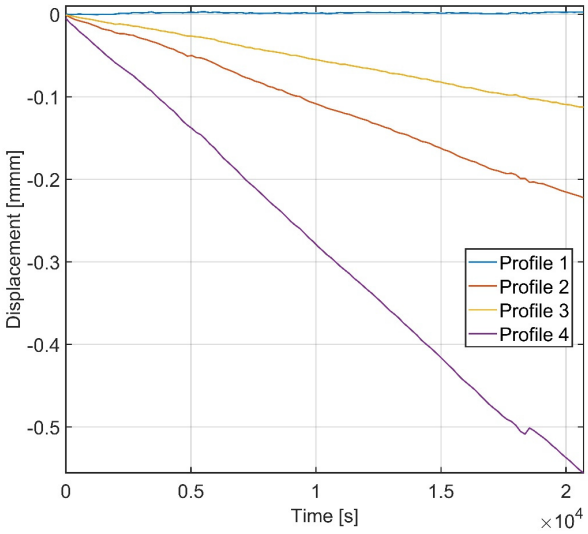


Fig. 3. Example of displacement profiles applied in the simulation.

TABLE II
NOMINAL AND RETRIEVED SPEEDS

Profile	Nominal speed (m/s)	Retrieved speed (m/s)
1	0	-7.18×10^{-9}
2	-0.56×10^{-5}	-0.556×10^{-5}
3	-1.11×10^{-5}	-1.12×10^{-5}
4	-2.78×10^{-5}	-2.79×10^{-5}

where v is the speed of the landslide, and $o(t^2)$ represents higher order acceleration terms.

B. APS Removal

The phasor z associated with the interferometric signal can be written as

$$z = e^{j\cdot(\phi_{\text{disp}}+\phi_{\text{APS}}+\phi_{\text{noise}})} = e^{j\left(\frac{4\pi}{\lambda}(v\,n\Delta t)+\frac{4\pi}{\lambda}(\delta n_o+\delta n_1 R)+\phi_{\text{noise}}\right)} \tag{6}$$

where Δt is the revisit time, and n represents the acquisition number. When large areas of the scene are affected by displacement, APS and displacement contributions are difficult to separate. However, only the displacement term has explicit temporal dependence. Therefore, the average velocity $\hat{v}$ can be estimated by computing the discrete fast Fourier transform of z over time

$$\begin{aligned}\text{FFT}(z) &= \sum_{n=0}^{N_t-1} z_n e^{-\frac{2\pi j}{N_t}kn} \\ &= e^{j\left(\frac{4\pi}{\lambda}(\delta n_o+\delta n_1 R)+\phi_{\text{noise}}\right)} \sum_{n=0}^{N_t-1} e^{j\frac{4\pi}{c}f_c(v\,n\cdot\Delta t)} e^{-\frac{2\pi j}{N_t}kn}\end{aligned} \tag{7}$$

where N_t is the number of temporal samples, c is the speed of light, f_c is the central frequency, and k is the frequency index. The estimated speed $\hat{v}$ is then obtained as

$$\hat{v} = \frac{c}{2f_c}\frac{k_{\max}}{T} - \frac{c}{4f_c}\frac{1}{\Delta t} = \frac{\lambda}{2}\frac{k_{\max}}{T} - \frac{\lambda}{4}\frac{1}{\Delta t} \tag{8}$$

where $k_{\max}$ is the frequency bin corresponding to the FFT peak, and $T = N_t \cdot \Delta t$ is the total observation interval. The corrective term accounts for negative velocities, and the resulting speed

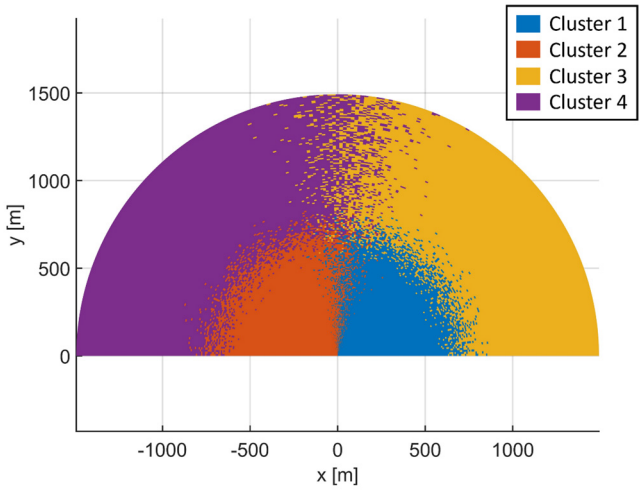


Fig. 4. Cluster map obtained from the simulated data.

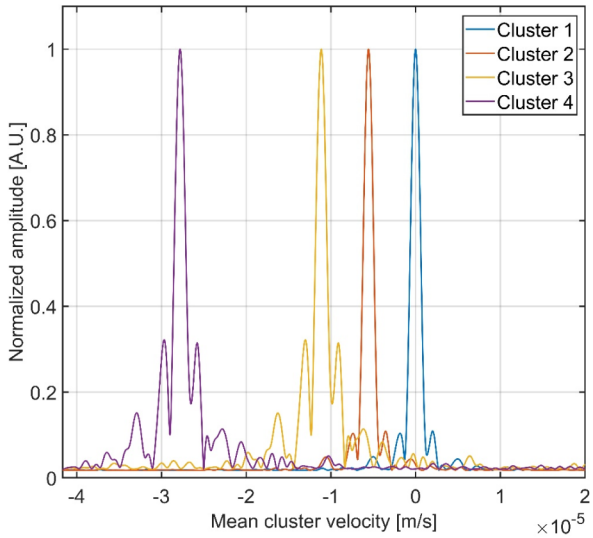


Fig. 5. Velocity spectra of the four clusters retrieved with (9).

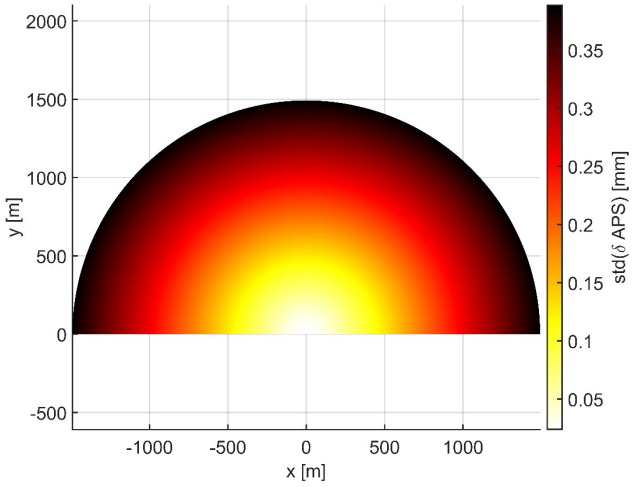


Fig. 6. Standard deviation of the difference between estimated and nominal APS.

domain is $(\pm\lambda/4 \cdot 1/\Delta t)$. In general, the Fourier transform requires a regularly sampled temporal baseline; however, the estimated velocity may remain reliable as long as no more than approximately 10% of the samples are missing.

By removing the estimated displacement contribution from (6), the APS component can be isolated

$$\hat{\boldsymbol{z}} = \boldsymbol{z} \cdot \left(\hat{\boldsymbol{z}}^{\boldsymbol{disp}}\right)^* \approx e^{j\left(\frac{4\pi}{\lambda}(\delta n_o + \delta n_1 R) + \phi_{\text{noise}}\right)} \tag{9}$$

where $\hat{\boldsymbol{z}}^{\boldsymbol{disp}}$ is the phasor corresponding to $\hat{v}$: $\hat{\boldsymbol{z}}^{\boldsymbol{disp}} = e^{j\frac{4\pi}{c} f_c (\hat{v} n \cdot \Delta t)}$. Any residual velocity is absorbed into the noise term.

Finally δn_0 and δn_1 can be estimated using a multiple PS (MPS) method [13], [17]. Once estimated, APS can be compensated, and the corrected interferometric signal is given by

$$\boldsymbol{z}_{\boldsymbol{corr}} = \boldsymbol{z} \cdot \left(e^{j\frac{4\pi}{\lambda}\left(\widehat{\delta n}_0 + \widehat{\delta n}_1 R\right)}\right)^* = e^{j\left(\frac{4\pi}{\lambda}\Delta R + \phi_{\text{noise}}\right)} \tag{10}$$

where $\widehat{\delta n}_0$ and $\widehat{\delta n}_1$ are the estimated APS coefficients.

III. Time-Series and Clustering

To improve the accuracy of velocity estimation, a sufficiently large dataset is required. Since temporal coherence limits the extension of the time series [5], the Fourier transform is computed in (7) over groups of pixels exhibiting similar phase behavior, so that $\hat{v}$ corresponds to the average velocity of the group.

The IP of the ith pixel can be stacked in a time vector

$$\boldsymbol{\phi}_i^T = \left[\phi_i^1, \phi_i^2, \ldots, \phi_i^M\right], \quad M = N_t - 1 \tag{11}$$

where ϕ_i^n is the IP at epoch t_n. In practice, it is often convenient to use the cumulative IP (CP)

$$\boldsymbol{\phi}_{i,\text{cum}}^T = \left[\phi_{i,\text{cum}}^1, \phi_{i,\text{cum}}^2, \ldots, \phi_{i,\text{cum}}^M\right] \tag{12}$$

with

$$\phi_{i,\text{cum}}^n = \phi_i^n + \phi_i^{n+1}. \tag{13}$$

The CP vector (12) is used for clustering. For the sake of brevity, we refer to CP, $\boldsymbol{\phi}_{i,\text{cum}}^T$, as $\boldsymbol{\phi}_i^T$.

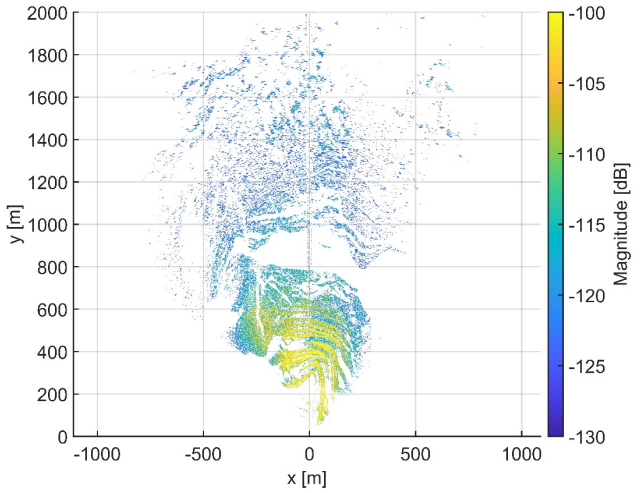


Fig. 7. Radar image of a copper mine in South America (DA < 0.35).

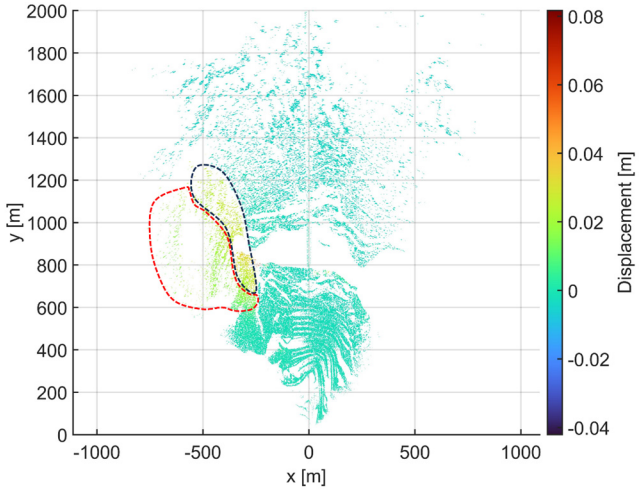


Fig. 8. Cumulative displacement map w/o APS compensation of a copper mine in South America.

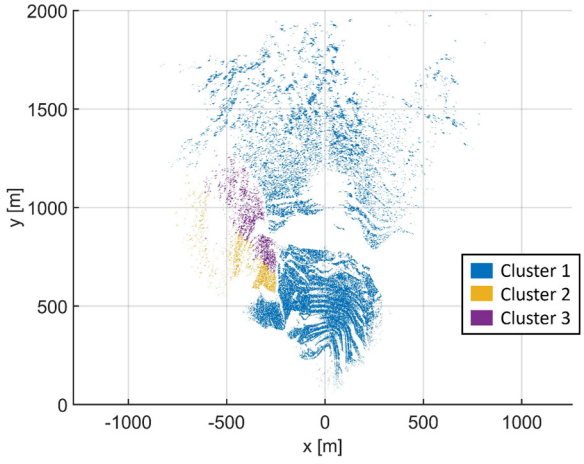


Fig. 9. Cluster map of a copper mine in South America.

To select a proper subset of coherent pixels, we compute the dispersion index (DA) of the pixel amplitude [5]

$$\text{DA}_i = \frac{\sigma_t\left(|I_i(t)|\right)}{\text{mean}_t\left(|I_i(t)|\right)} \tag{14}$$

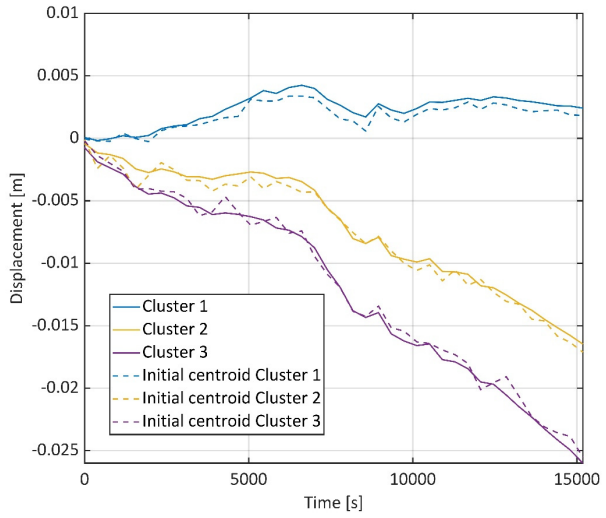


Fig. 10. Cluster-average displacement [from (4)] (copper mine).

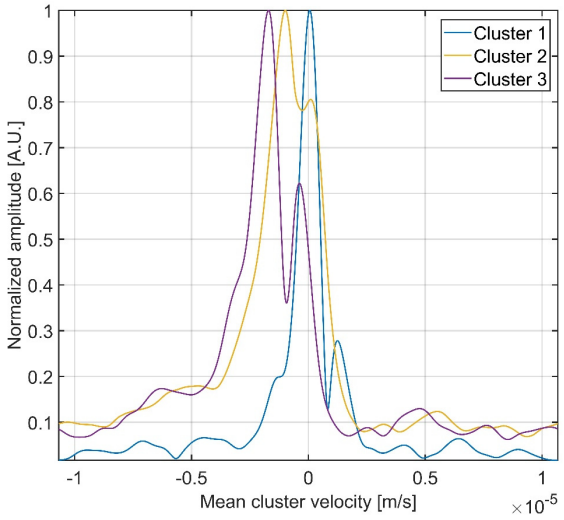


Fig. 11. Cluster velocity spectra from (9) (copper mine).

TABLE III
AVERAGE SPEED (COPPER MINE)

Cluster	Speed (m/s)
1	$\mathbf{6.41 \times 10^{-8}}$
2	-9.62×10^{-8}
	$\mathbf{-9.94 \times 10^{-7}}$
3	-3.85×10^{-7}
	$\mathbf{-1.71 \times 10^{-6}}$

where $\sigma_t(\cdot)$ is the temporal standard deviation (STD). Pixels with DA $< 0.35 - 0.4$ are classified as permanent scatterers (PSs).

Then, k-means clustering is applied to group PSs with similar temporal signatures. The k-means clustering iteratively partitions the dataset into k clusters by minimizing the distance metric

$$d\left(\boldsymbol{\phi}_i, \boldsymbol{\mu}_k\right) = \sqrt{\sum\nolimits_n \left(\phi_i^n - \mu_k^n\right)^2} \times \frac{\sqrt{\left(x_i - \bar{y}_k\right)^2 + \left(y_i - \bar{y}_k\right)^2}}{R_{\max} - R_{\min}} \tag{15}$$

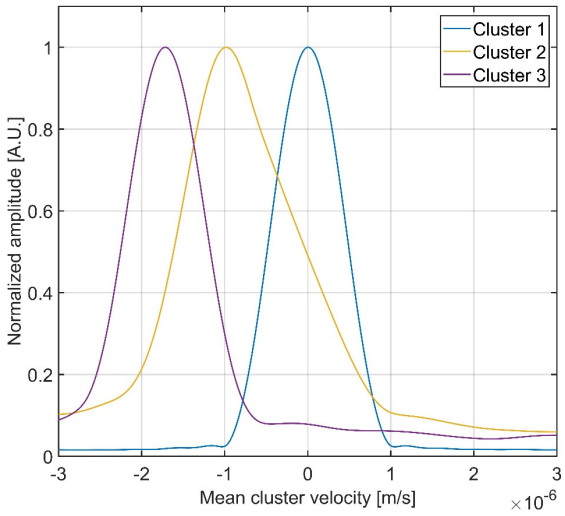


Fig. 12. Velocity spectra after APS compensation (copper mine).

TABLE IV
RETRIEVED SPEEDS BEFORE/AFTER APS (COPPER MINE)

Profile	Speed w APS (m/s)	Speed w/o APS (m/s)
1	6.41×10^{-8}	0
2	-9.62×10^{-8}	ND
	-9.94×10^{-7}	-9.83×10^{-7}
3	-3.85×10^{-7}	ND
	-1.71×10^{-6}	-1.72×10^{-6}

where μ_k is the cluster centroid in phase space, (x_i, y_i) are the coordinates of pixel i, and $(\bar{x}_k, \bar{y}_k)$ are the cluster barycentre coordinates. This metric accounts for both phase similarity and spatial proximity.

The proposed clustering algorithm proceeds as follows.

1) Define the number of clusters and initialize each cluster centroid (μ_k) with the CP of a selected PS.

2) Initialize cluster barycentre with the selected PS coordinates.

3) Compute the distance metric (15) for each PS to all clusters.

4) Assign the PS to the cluster that minimizes (15).

5) Update μ_k and the cluster centroid and barycentre as follows:

$$\mu_k^n = \text{mean}_i\left(\phi_i^n\right)$$
$$\bar{x}_k = \text{mean}\left(x_i^n\right), \bar{y}_k = \text{mean}\left(y_i^n\right), \quad \text{for } i \in k\text{-cluster}.$$

6) Repeat steps 3–5 until cluster barycentre converges within a predefined threshold.

IV. SIMULATIONS

To validate the proposed APS removal and clustering algorithm, a radar operating at a central frequency of $f_c = 10$ GHz with a repeat time of 180 s is simulated. The simulation spans 5.85 h, corresponding to $N_t = 117$ acquisitions. Each image consists of 200×200 pixels divided into four areas with different displacement profiles, as shown in Fig. 1. To reproduce a realistic scenario, gradual transitions between

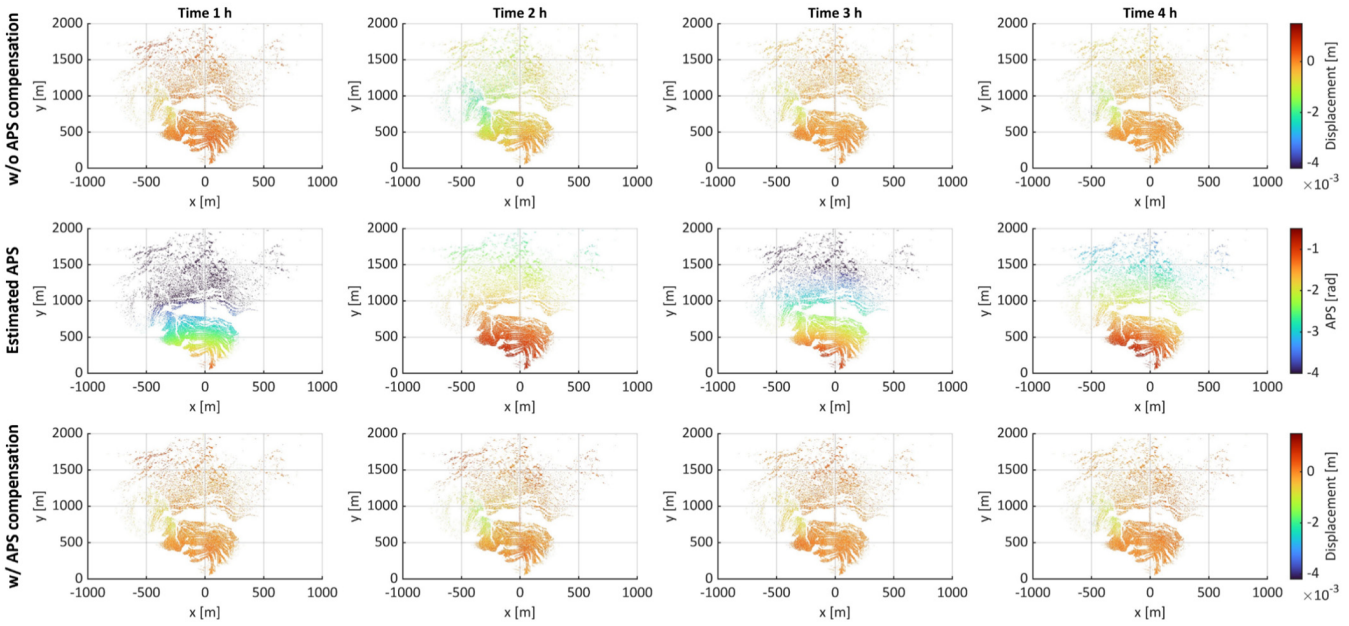


Fig. 13. Interferograms before/after APS correction and estimated APS at hours 1, 2, 3, and 4 (copper mine).

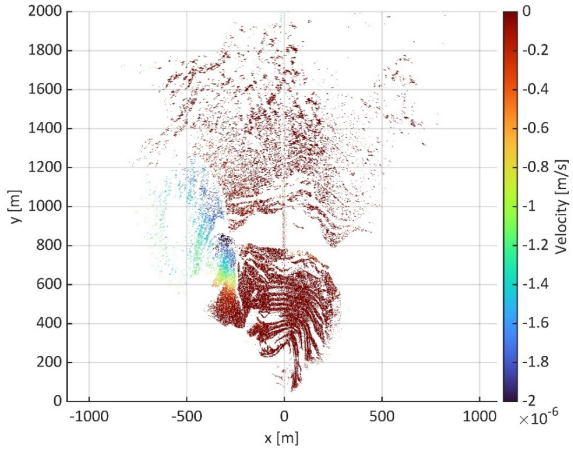


Fig. 14. Velocity map after APS compensation (copper mine).

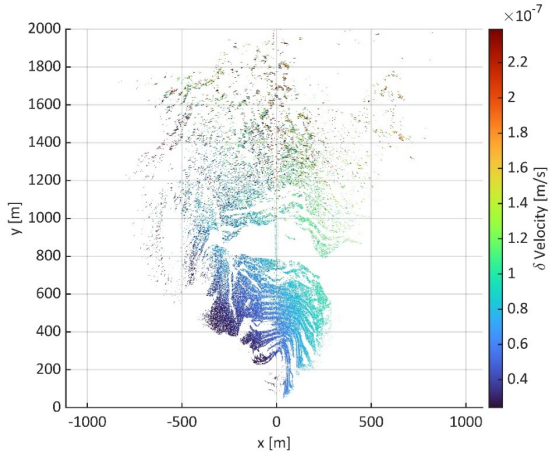


Fig. 15. Map of the difference in velocity before and after APS compensation (copper mine).

areas are introduced. The average velocities of the four profiles are reported in Table I. The APS is applied to the simulated images according to (3), using a phase screen extracted from a real measurement campaign (Fig. 2). Finally, Gaussian noise with a STD of 20 mm is added to each image. An example of the displacement profiles is shown in Fig. 3.

The cluster map obtained from the simulation is shown in Fig. 4. The algorithm successfully detects the four clusters. Fig. 5 shows the average velocity spectra, obtained by averaging the spectra of (9) over all pixels in each cluster. The retrieved velocities, corresponding to the spectral peaks, are reported in Table II. A good agreement between the nominal and retrieved speeds is observed. For clusters 3 and 4, a small discrepancy of the order of 10^{-7} m/s is present, due to stronger APS effects from (3), as these clusters are located at larger ranges.

Finally, Fig. 6 shows the STD between the estimated APS and the nominal APS (δ APS). The STD is between 0.06 and 0.39 mm and depends only on range. It is important to note that no dependency on the cluster is observed.

V. Experimental Results

The proposed APS-removal and clustering algorithm was validated on two experimental datasets: 1) an open-pit copper mine in South America and 2) the Belvedere Glacier on the east face of Monte Rosa.

A. Open-Pit Copper Mine

Open-pit monitoring is among the most common GBSAR applications. The method was tested on a representative

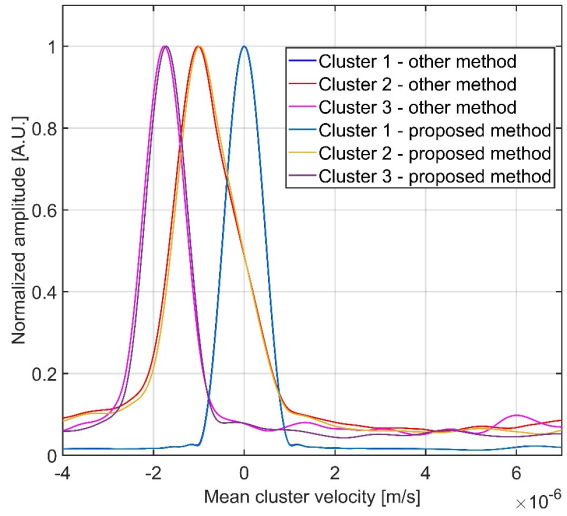


Fig. 16. Velocity spectra comparison: proposed APS versus MRL-APS (copper mine).

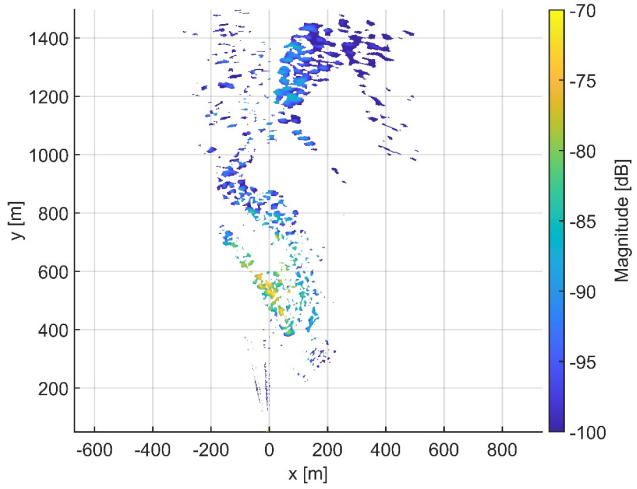


Fig. 17. Radar image of the Belvedere Glacier, Italy (DA < 0.4).

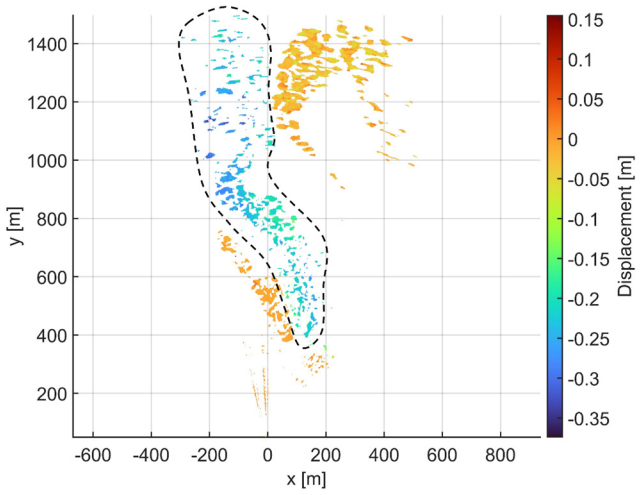


Fig. 18. Cumulative displacement map w/o APS compensation of the Belvedere glacier.

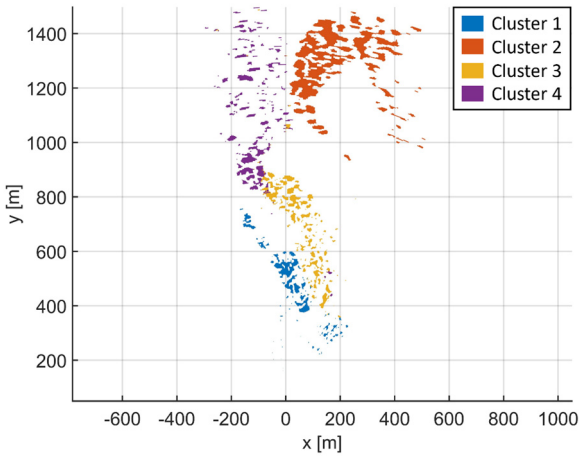


Fig. 19. Cluster map of the Belvedere glacier.

copper-mine dataset acquired with the following parameters: initial frequency $f_1 = 17.1$ GHz, final frequency $f_2 = 17.3$ GHz, number of frequencies $N_f = 5333$, mechanical scan length $L = 1.275$ m, with number of spatial samples $N = 256$, repeat time of 389.35 s, and 41 observations over almost 4 h. The radar amplitude image is shown in Fig. 7, in which the PSs are selected with DA < 0.35.

The cumulative displacement map without APS compensation is shown in Fig. 8. Three main areas can be identified: the regions highlighted in blue and red are affected by movements at two different rates, while the remaining area is essentially stable.

According to Fig. 8, three clusters were initialized. The resulting cluster map is shown in Fig. 9, while Fig. 10 reports the cluster-average phase [converted to displacement via (4)] with a comparison with initial centroid displacement. A slight trend is also visible for cluster 1, which is expected to be quasi-stationary.

The average velocity spectra [obtained by averaging the spectra of (9) over all pixels in each cluster] are shown in Fig. 11. For clusters 2 and 3, two peaks are present: the main peak represents the average velocity, whereas the secondary peak is attributable to a residual APS trend over time, consistent with the 4-h acquisition window during which APS variations are not fully random. The estimated speeds are summarized in Table III; values used for APS removal are highlighted in bold.

TABLE V
AVERAGE SPEEDS (BELVEDERE GLACIER)

Cluster	Speed (m/s)
1	2.38×10^{-8}
2	-1.60×10^{-7}
	8.81×10^{-8}
	$\mathbf{4.00 \times 10^{-7}}$
3	-9.36×10^{-7}
4	-1.10×10^{-6}

After APS compensation, the secondary peaks disappear (Fig. 12), confirming their APS origin. Table IV reports the

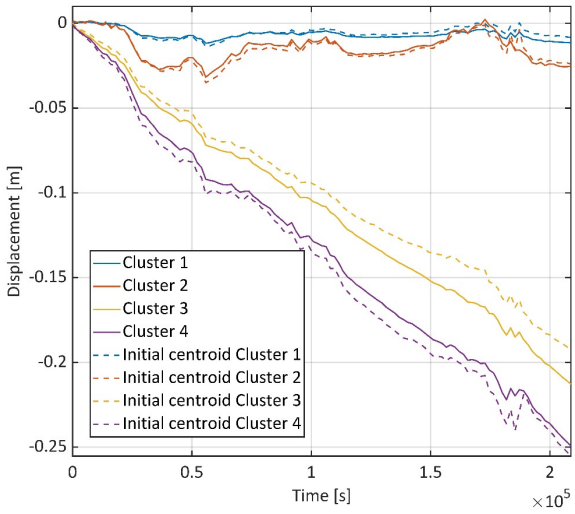


Fig. 20. Cluster-average displacement [from (4)] (Belvedere glacier).

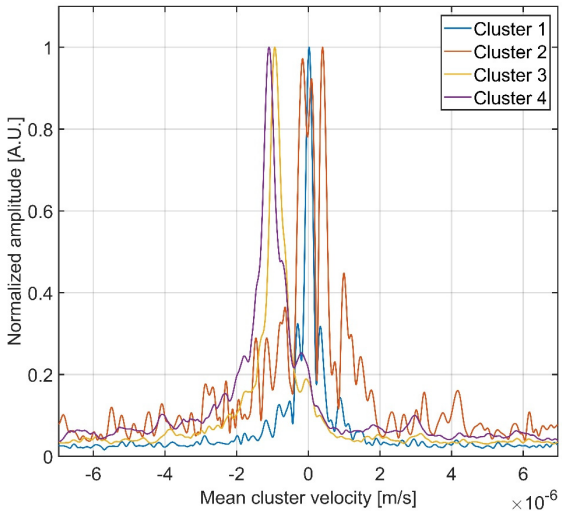


Fig. 21. Cluster velocity spectra from (9) (Belvedere glacier).

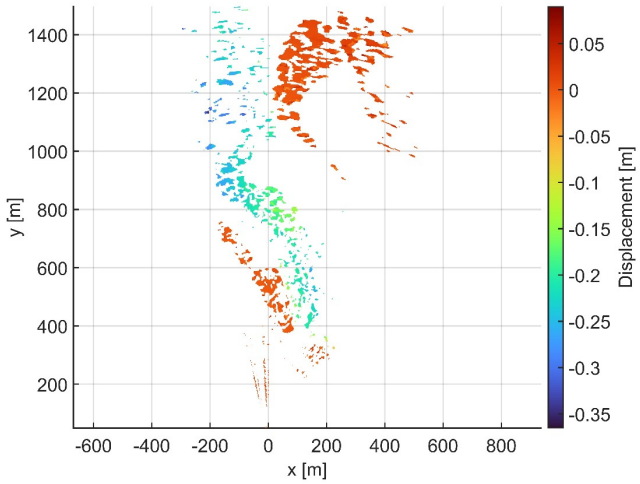


Fig. 22. Cumulative displacement map w/APS compensation (Belvedere glacier).

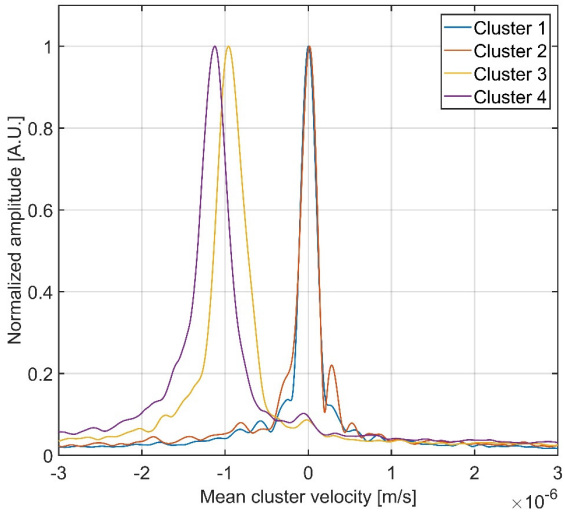


Fig. 23. Velocity spectra after APS compensation (Belvedere glacier).

average velocities before/after APS removal; the agreement is good, with a maximum discrepancy of 8 ‰ for cluster 1.

Fig. 13 shows the interferograms before and after APS correction, together with the estimated APS, for the open-pit scenario. The interferograms were computed between two consecutive measurements. Four interferograms were randomly selected from the first, second, third, and fourth h of acquisition. As expected for this scenario, the APS varies significantly between the different image pairs because of rapid atmospheric fluctuations. After correction, the interferograms become highly consistent across time.

The velocity map after APS compensation is shown in Fig. 14. The PSs corresponding to cluster 1 in Fig. 9 have a speed equal to zero. The moving areas (cluster 2–3) exhibit a significant spatial gradient in deformation, with a velocity variation of approximately 4×10^{-7} m/s over 100 m. Based on these results, the proposed method appears to be robust for a spatial gradient of velocity (or at least for a gradient of this magnitude).

Fig. 15 shows the difference in velocity before and after APS compensation. The difference is of the order of 10^{-7} m/s, which is compatible with the difference in velocity spectra.

For comparison, a model-based APS (MPR-APS [13], [17]) using only PSs with positive x (Fig. 8 for reference), assumed motionless, was applied. The APS estimated from half the image was applied to the entire scene, and the velocity spectra were recomputed using the clusters of Fig. 9.

As shown in Fig. 16, cluster-1 spectra (nonmoving PSs) are nearly identical, whereas for clusters 2 and 3, the MRP-APS average velocities have a discrepancy of about 2%–3%.

B. Glacier

As a natural test site, the Belvedere Glacier (Italian Alps) was considered, using data from a 2006 campaign described in [26]. Acquisition parameters were: $f_1 = 5.97$ GHz, $f_2 = 5.99$ GHz, $Nf = 801$, $L = 1.71$ m, $N = 141$, repeat time of 1800 s and 118 observations for a total time of ~2.5 days. 9:08 AM The radar image is shown in Fig. 17 with a PS selection with a threshold of DA < 0.4.

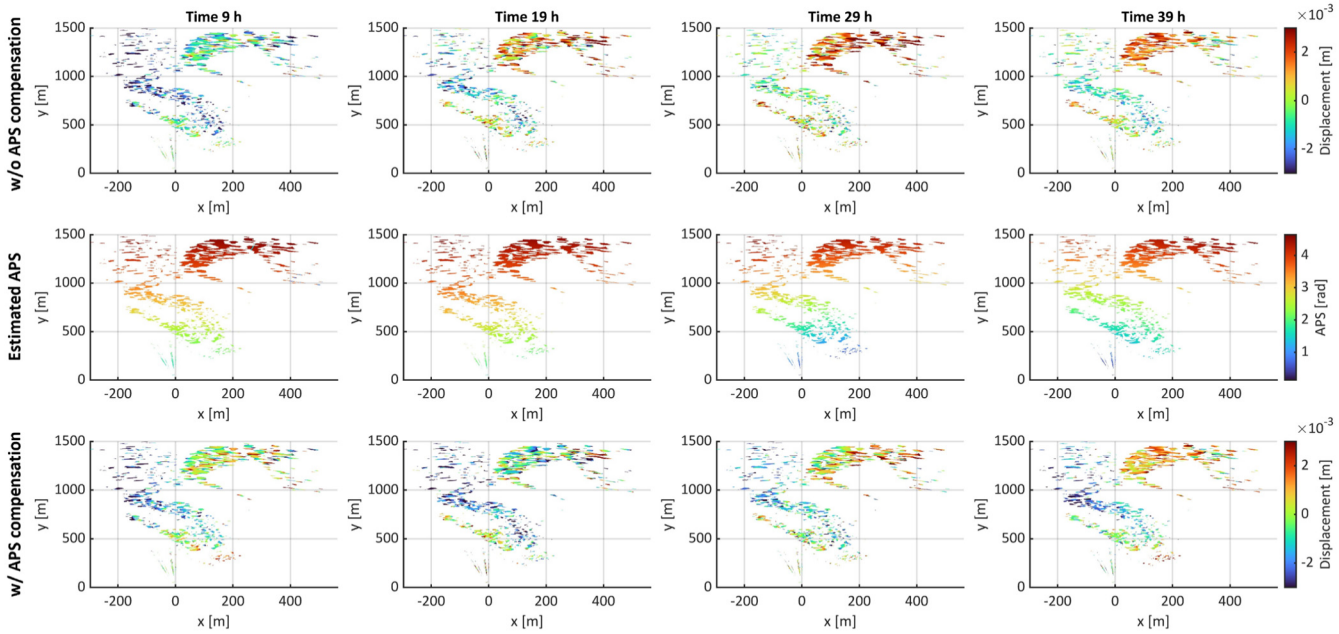


Fig. 24. Interferograms before/after APS correction and estimated APS at hours 9, 19, 29, and 39 (Belvedere glacier).

TABLE VI
RETRIEVED SPEEDS BEFORE/AFTER APS (BELVEDERE GLACIER)

Profile	Speed w APS (m/s)	Speed w/o APS (m/s)
1	2.38×10^{-8}	2.38×10^{-9}
2	-1.60×10^{-7}	
	8.81×10^{-8}	1.42×10^{-8}
	4.00×10^{-7}	
3	-9.36×10^{-7}	-9.88×10^{-7}
4	-1.10×10^{-6}	-1.12×10^{-6}

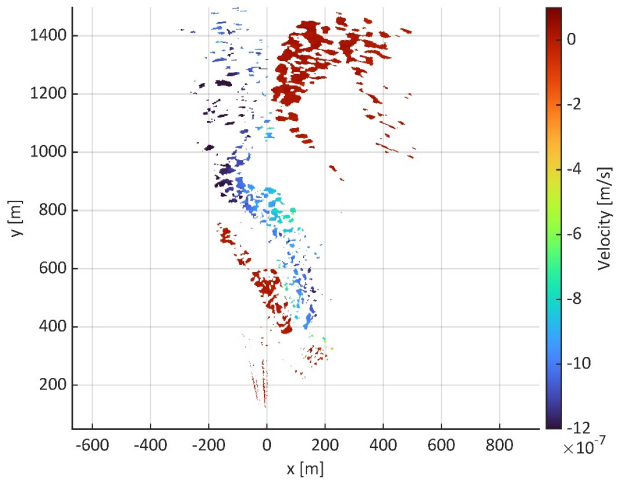


Fig. 25. Velocity map after APS compensation (Belvedere glacier).

This scene includes a wide area undergoing displacement (dotted outline in Fig. 18). In such conditions, classic MPR-APS methods are ineffective because most PSs are moving. A manual APS fit estimation based on a few hand-picked stationary PSs is possible but statistically weak. In contrast, the proposed clustering-based APS compensation increases the pixel statistics for APS estimation.

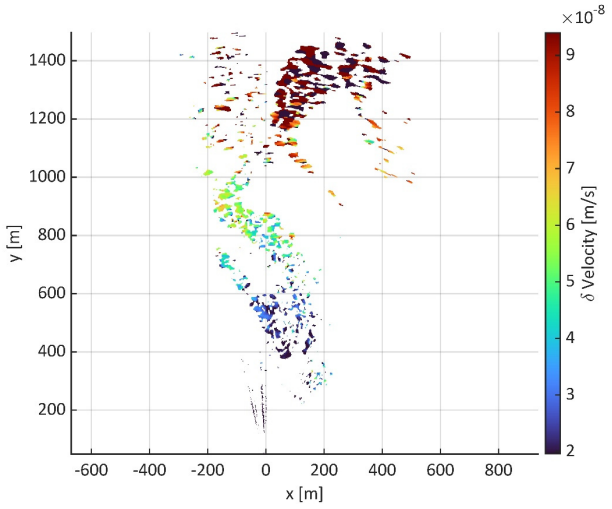


Fig. 26. Map of the difference in velocity before and after APS compensation (Belvedere glacier).

Based on prior knowledge, four clusters were initialized (Fig. 19). Clusters 1–2 are quasi-stationary, while clusters 3–4 show distinct displacement behaviors (Fig. 20). The comparison with the initial centroid and average cluster displacement is shown in Fig. 20.

The velocity spectra are shown in Fig. 21 and the peak values in Table V. Clusters 1, 3, and 4 exhibit a single dominant peak (coherent motion). Cluster 2 shows three peaks: two with appreciable speeds and one near zero, likely due to the lower SNR; as discussed below, cluster 2 is effectively stationary. The main peak in each cluster was used for APS compensation.

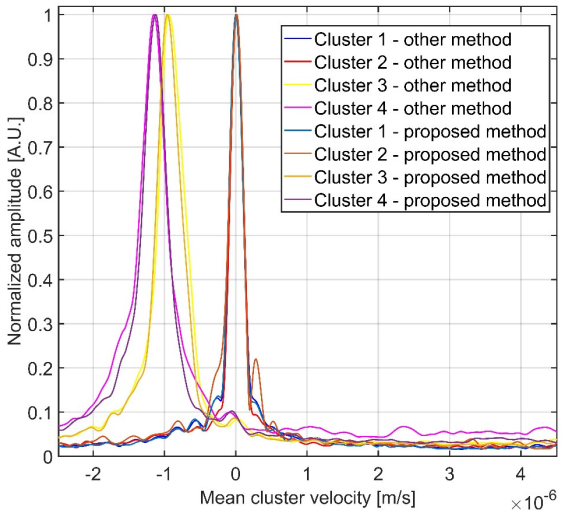


Fig. 27. Velocity spectra comparison: proposed APS versus linear APS fit (Belvedere glacier).

After APS compensation, the cumulative displacement map (Fig. 22) shows that clusters 1–2 are stationary, while clusters 3–4 undergo displacements consistent with the glacier motion. The post-compensation spectra (Fig. 23) show that the secondary peaks in cluster 2 vanish. Table VI compares the before/after APS velocities, confirming good agreement.

The interferograms before and after APS compensation, together with the estimated APS, are shown in Fig. 24. The SNR of this scenario is lower than that of the copper mine, and the retrieved interferograms appear less homogeneous with respect to the previous case, in particular for the farthest PSs, which correspond to cluster 2 in Fig. 19.

Fig. 25 shows the velocity map after APS compensation. The speed profile corresponds to the clustering in Fig. 19, with two areas almost stationary and two areas with different speeds. The difference in the velocity map before and after correction is shown in Fig. 26. In this case, the average difference is lower than in the previous case.

Finally, for comparison, a simple linear APS fit using PSs from clusters 1 and 2 only was applied. The modeled APS is propagated across the whole scene, and spectra are recomputed using the clusters of Fig. 19. Results in Fig. 27 show a discrepancy between peak velocities of about 2.5%–4.5%.

VI. Conclusion

In this article, the authors presented a spectral-analysis-based approach for APS removal in GB-SAR interferometry. Unlike traditional model-based APS estimation methods, which rely on the assumption of a predominance of stationary pixels, the proposed method can be used for scenarios with large areas affected by displacement. Fourier transform of the IP is calculated to retrieve an average velocity of each pixel of the image. This velocity contribution is subtracted in the complex domain, enabling all pixels to be virtually motionless and thus available for APS estimation. To further improve robustness, a k-means clustering strategy was introduced to group pixels with similar temporal signatures, thereby increasing the SNR and enhancing the accuracy of velocity retrieval. A key advantage of the proposed framework is that APS estimation can be performed over the entire scene, without the need for manually identifying static areas. This makes the approach well suited for scenarios where large portions of the scene are affected by deformation (e.g., glaciers) and enhances automation, processing speed, and operator independence.

The proposed algorithm was first validated through simulations, showing good agreement between nominal and retrieved velocities, with APS estimation errors of the order of submillimetric STD. The method was then tested on two experimental datasets: an open-pit copper mine in South America and the Belvedere Glacier, Italy. In both cases, the proposed APS correction effectively reduced residual atmospheric effects, leading to displacement estimates consistent with the expected motion patterns. The copper mine scenario presents a deformation gradient of about 4×10^{-7} m/s over 100 m. The proposed method is robust for speed gradient of at least this magnitude. For scenarios characterized by stronger spatial gradients, the number of clusters should be increased to ensure that each cluster contains pixels with sufficiently homogeneous phase behavior. The method remains reliable as long as each cluster includes enough pixels to compute a stable average velocity.

The comparison with model-based solutions based on MPS confirmed the improved accuracy and stability of the approach, especially in scenarios where a large fraction of the scene is affected by displacement.

The proposed method presents some inherent limitations. A major assumption is that the displacement evolves approximately linearly in time; in the presence of higher order motion components (e.g., acceleration), the velocity spectra may exhibit multiple peaks, reducing the accuracy of the estimated average speed. The robustness of the velocity estimation also depends on the number of coherent PSs within each cluster: in low-SNR conditions or when few PSs are available, complementary approaches—such as DS-oriented methods—may be employed to increase statistical reliability. Furthermore, the method requires a sufficiently long time series to ensure adequate spectral resolution. The time series should be acquired with a regularly sampled temporal baseline; however, the analysis performed indicates that the velocity estimate remains reliable if the percentage of missing samples does not exceed roughly 10%.

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